House Price Prediction Using Machine Learning

1. Objective

The objective of this project is to develop a machine learning model capable of accurately predicting residential house prices based on structural, geographical, and neighborhood characteristics. Multiple regression algorithms were implemented and compared to identify the best-performing model.

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2. Dataset

Dataset: King County House Sales Dataset

- Number of observations: **21,613**
- Number of features: **21**
- Target Variable:
  - **price**

The dataset contains information regarding house characteristics such as:

- Bedrooms
- Bathrooms
- Living Area
- Lot Area
- Floors
- Grade
- Condition
- Waterfront
- View
- Latitude
- Longitude
- Year Built
- Year Renovated

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3. Data Preprocessing

The following preprocessing steps were performed.

Data Cleaning

- Removed the **id** column since it has no predictive value.
- Converted the **date** column into datetime format.
- Removed the **zipcode** column.
- Checked for missing values.
- Checked for duplicate observations.

Result:

- Missing Values: **0**
- Duplicate Values: **0**

Outlier Analysis

Boxplots were generated for all numerical variables.

An observation containing **33 bedrooms** was identified as an obvious data-entry error.

This observation was removed from the dataset.

All remaining extreme observations were retained because they represented genuine luxury houses rather than erroneous records.

Target Variable Transformation

The distribution of house prices exhibited severe positive skewness.

Original Skewness:

4.024

A logarithmic transformation was applied:

price_log = log1p(price)

New Skewness:

0.428

The transformed target showed a nearly symmetric distribution, making it more suitable for regression modeling.

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4. Exploratory Data Analysis

Several visualization techniques were used.

Distribution Analysis

- Histogram of house prices
- Histogram after logarithmic transformation

Observation:

- Original prices were heavily right-skewed.
- Log transformation significantly normalized the distribution.

Pair Plot Analysis

The following variables showed strong relationships with price:

- sqft_living
- grade
- bathrooms
- sqft_above

Living area exhibited the strongest positive relationship with house prices.

Correlation Analysis

A correlation matrix was generated.

Highly correlated variables included:

- sqft_living
- sqft_above
- sqft_basement

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5. Multicollinearity Analysis

Variance Inflation Factor (VIF) was calculated.

Initially,

sqft_living
sqft_above
sqft_basement

produced infinite VIF values due to the relationship

sqft_living = sqft_above + sqft_basement

To remove perfect multicollinearity,

sqft_above
sqft_basement

were removed.

After recalculating VIF:

Maximum VIF

sqft_living = 5.06

which is considered acceptable.

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6. Model Development

The dataset was divided into

- Training Set (80%)
- Testing Set (20%)

Pipelines were implemented to prevent data leakage.

Feature scaling was applied only where necessary.

Models Implemented

1. Linear Regression

Pipeline:

- StandardScaler
- Linear Regression

Performance

- MAE: **112,901**
- RMSE: **220,039**
- R^2: **0.677**

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2. Ridge Regression

Pipeline:

- StandardScaler
- Ridge Regression

Performance

- MAE: **112,902**
- RMSE: **220,050**
- R^2: **0.677**

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3. Lasso Regression

Pipeline:

- StandardScaler
- Lasso Regression

Performance

- MAE: **112,896**
- RMSE: **221,706**
- R^2: **0.673**

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4. Random Forest Regressor

Pipeline:

- Random Forest Regressor

Performance

- MAE: **73,305**
- RMSE: **140,413**
- R^2: **0.869**
- MAPE: **12.97%**

Observation:

Random Forest captured the complex nonlinear relationships between housing features and selling price much better than linear models.

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5. XGBoost Regressor

Pipeline:

- XGBRegressor

Performance

- Training R^2: **0.9455**
- Testing R^2: **0.9090**

Difference

0.0365

The small gap between training and testing performance indicates that the model generalized well and did not exhibit significant overfitting.

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7. Model Comparison

Model	MAE	RMSE	R^2
Linear Regression	112,901	220,039	0.677
Ridge Regression	112,902	220,050	0.677
Lasso Regression	112,896	221,706	0.673
Random Forest	**73,305**	**140,413**	**0.869**
XGBoost	*Add final MAE/RMSE after evaluation*	*Add*	**0.909**

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8. Key Findings

- Living area was the strongest predictor of house prices.
- Log transformation significantly improved the distribution of the target variable.
- Perfect multicollinearity was removed by eliminating redundant features.

- Linear models performed reasonably well but struggled to model nonlinear relationships.
- Random Forest substantially reduced prediction error.
- XGBoost achieved the highest predictive performance with minimal overfitting.

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9. Conclusion

This project demonstrated the complete workflow of a supervised machine learning regression problem, including data preprocessing, exploratory data analysis, multicollinearity assessment, feature engineering, model development, and evaluation.

Among all the evaluated algorithms, **XGBoost** achieved the best predictive performance (**R^2 = 0.909**), followed by **Random Forest (R^2 = 0.869)**. The train-test performance gap for XGBoost was only **0.0365**, indicating good generalization.

The study also highlighted the importance of proper preprocessing—particularly log transformation of the target variable and removal of perfect multicollinearity—in improving model performance.

Future Scope

- Hyperparameter tuning using GridSearchCV or RandomizedSearchCV.
- Incorporate additional location-based features or socioeconomic indicators.
- Explore advanced ensemble methods such as LightGBM and CatBoost.
- Deploy the best-performing model as a web application using Flask or Streamlit.
- Apply model interpretability techniques such as SHAP to explain individual predictions and feature contributions.

Overall, this is a project that demonstrates a complete machine learning workflow—from data cleaning and exploratory analysis to comparing multiple regression algorithms—and is well suited for inclusion in a portfolio or discussion during placements.